

1. What is a hypothesis?

- A list of materials used in a lab.
- The final score of an experiment.
- A proven fact that never changes.
- A testable prediction, often written as an 'If..., then...' sentence.

Answer:

2. What is the independent variable in an experiment?

- The variable that is measured at the end.
- The one variable that the scientist changes on purpose.
- The equipment used to measure data.
- A variable that stays the same to keep the test fair.

Answer:

3. In the parachute experiment, what is the dependent variable?

- The time it takes for the parachute to fall to the floor.
- The color of the plastic sheet.
- The size of the parachute canopy.
- The weight of the paperclip.

Answer:

4. Which of the following is an investigation question?

- The hypothesis was supported by our results.
- We measured 150 mL of water in a glass beaker.
- How does water temperature affect the time it takes for sugar to dissolve?
- Sugar dissolves faster in hot water than in cold water.

Answer:

5. What is the dependent variable in an experiment?

- A factor kept exactly the same in all trials.
- A rule in the lab to keep students safe.
- The variable that the scientist decides to change.
- The variable that is measured or observed as the result.

Answer:

6. In a lab report, what do you do in the 'Analysis' section?

- Guess what might happen before doing any tests.
- Look closely at the data to find patterns and compare results.
- Clean up the lab table and put away tools.
- Choose a new topic to study.

Answer:

7. Why do scientists repeat an experiment multiple times (doing several trials)?

- To use up all the leftover lab materials.
- To change the independent variable each time.
- To get reliable results and calculate an accurate average.
- To make the experiment more confusing.

Answer:

8. In the ramp experiment, the ramp height was tested at 5 cm, 10 cm, 15 cm, and 20 cm. The ramp height is the:

- Controlled variable.
- Independent variable.
- Control group.
- Dependent variable.

Answer:

9. What is the main purpose of the 'Conclusion'?

- To list the names of everyone in the group.
- To state whether the results support the hypothesis and explain what was learned.
- To measure the temperature of water.
- To write down the step-by-step procedure.

Answer:

10. Why are controlled variables important in an experiment?

- They are the main result we want to measure.
- They change randomly during every trial.
- They make sure the experiment is a fair test.
- They make the experiment finish faster.

Answer:
